THERMODYNAMIC FORMALISM MEETS ERGODIC OPTIMIZATION

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This note presents a concrete example illustrating the following connection between the thermodynamic formalism and ergodic optimization, in the context of MANNEVILLE–POMEAU maps and HÖLDER continuous potentials:

(*) *If a potential undergoes a phase transition in temperature, then the* DIRAC *measure at the neutral fixed point uniquely maximizes the potential.*

This result is a direct consequence of [CIR25, Key Lemma and Proposition 2.10]. For background on ergodic optimization, see, for example, [Jen19].

The example relies on the following notation and terminology. Fix α in $(0, +\infty)$ and the MANNEVILLE–POMEAU or *intermittent* map $f\colon [0,1] \to [0,1]$ defined by

$$f(x) := \begin{cases} x(1 + x^{\alpha}) & \text{if } x(1+x^{\alpha}) \leqslant 1; \\ x(1+x^{\alpha}) - 1 & \text{otherwise.} \end{cases} \tag{1}$$

Denote by $\mathscr{M}$ the space of BOREL probability measures on $[0,1]$ that are invariant by f and, for each ν in $\mathscr{M}$, by h_{ν} the measure-theoretic entropy of ν. For each continuous function $\varphi\colon [0,1] \to \mathbb{R}$, the *pressure* $P(\varphi)$ *of* f *for the potential* φ is defined by

$$P(\varphi) := \sup \left\{ h_{\nu} + \int \varphi \, \mathrm{d}\nu \colon \nu \in \mathscr{M} \right\}. \tag{2}$$

Proposition 1. *Let γ in $(0, +\infty)$ be given, and define the potential $\widehat{\varphi}\colon [0,1] \to \mathbb{R}$ by*

$$\widehat{\varphi}(x) := -x^{\gamma}(1-x). \tag{3}$$

Then, $t \mapsto P(t\widehat{\varphi})$ is real-analytic on $(0, +\infty)$.

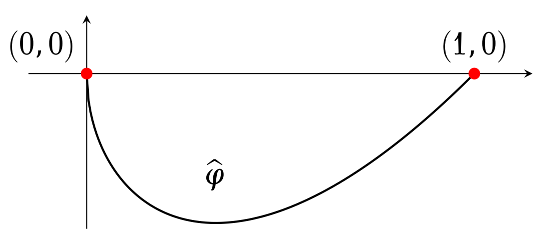

FIGURE 1. The graph of the potential $\widehat{\varphi}$ with $\gamma = \frac{1}{2}$.

The potential $\widehat{\varphi}$ is nonpositive and vanishes precisely at the neutral fixed point 0 and at the repelling fixed point 1. Thus, the measure δ_0 maximizes $\widehat{\varphi}$, but not uniquely. The conclusion of Proposition 1 is that the potential $\widehat{\varphi}$ does not undergo a phase transition in temperature, as ensured by (*).

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This example is [CIR25, Example 1.2]. The remainder of this note provides a streamlined proof of Proposition 1. The proof relies on a few lemmas. Denote by x_1 the unique discontinuity point of f.

Lemma 2. *For every β in $(0, +\infty)$,*

$$\sup_{[0,1]} \beta\widehat{\varphi} < P(\beta\widehat{\varphi}). \tag{4}$$

In particular, the potential $\beta\widehat{\varphi}$ is hyperbolic for f in the sense of [IRRL12].

Proof. Denote by x_2 the unique preimage of x_1 by f in $(0, x_1)$, and by X the maximal invariant set of f in $[x_2, 1]$. Then, $f|_X$ is uniformly expanding and topologically conjugate to the golden mean shift, via a Hölder continuous conjugacy. By the thermodynamic formalism for Hölder continuous potentials, it follows that $f|_X$ has a unique Gibbs state for the potential $\beta\widehat{\varphi}|_X$ and that the measure-theoretic entropy of this measure is strictly positive. See, for example, [Bow08, Theorem 1.25]. It follows that δ_1 is not an equilibrium state of $f|_X$ for the potential $\beta\widehat{\varphi}|_X$, and thus

$$P(\beta\widehat{\varphi}) \geqslant P_{f|_X}(\beta\widehat{\varphi}|_X) > h_{\delta_1}(f) + \int \beta\widehat{\varphi}|_X \, \mathrm{d}\delta_1 = 0. \tag{5}$$

Since $\widehat{\varphi} \leqslant 0$ on $[0, 1]$, the inequality (4) follows. $\square$

Lemma 3. *Let β be in $(0, +\infty)$ and put $g := \exp(\beta\widehat{\varphi} - P(\beta\widehat{\varphi}))$. Then, there is a $(1/g)$-conformal probability measure m for f that is atom-free and satisfies $\operatorname{supp}(m) = [0, 1]$.*

The proof of this lemma relies on the following one. Put $\Sigma := \{0, 1\}^{\mathbb{N}_0}$ and endow it with the distance dist defined, for distinct $(x_n)_{n=0}^{+\infty}$ and $(x'_n)_{n=0}^{+\infty}$ in Σ, by

$$\operatorname{dist}((x_n)_{n=0}^{+\infty}, (x'_n)_{n=0}^{+\infty}) := 2^{-\min\{n : \, n \in \mathbb{N}_0, x_n \neq x'_n\}}. \tag{6}$$

The *shift map* $\sigma \colon \Sigma \to \Sigma$ is defined by $\sigma\left((x_n)_{n=0}^{+\infty}\right) := (x_{n+1})_{n=0}^{+\infty}$. Denote by $C(\Sigma)$ the Banach space of real-valued continuous functions defined on Σ endowed with the supremum norm $\|\cdot\|_\Sigma$, and by $\mathbb{1}_\Sigma$ the function in $C(\Sigma)$ that is constant equal to 1. For ϕ in $C(\Sigma)$ denote by $P_\sigma(\phi)$ the pressure of σ for the potential φ. The *transfer operator $\mathcal{L}_{\sigma,\phi}$ of σ for the potential ϕ* is defined, for ψ in $C(\Sigma)$, by

$$\mathcal{L}_{\sigma,\phi}\psi(\underline{x}) := \sum_{\underline{x}' \in \sigma^{-1}(\underline{x})} \exp(\phi(\underline{x}'))\psi(\underline{x}'). \tag{7}$$

Lemma 4. *For every ϕ in $C(\Sigma)$, the sequence $\left(\frac{1}{n} \log \mathcal{L}_{\sigma,\phi}^n \mathbb{1}_\Sigma\right)_{n=1}^{+\infty}$ converges uniformly to $P_\sigma(\phi) \cdot \mathbb{1}_\Sigma$.*

Proof. For all n in $\mathbb{N}$ and $x_0 \ldots x_{n-1}$ in $\{0, 1\}$, the *cylinder* $[x_0 \ldots x_{n-1}]$ is defined by

$$[x_0 \ldots x_{n-1}] := \{(x'_n)_{n=0}^{+\infty} \in \Sigma : \, x'_m = x_m, m \in \{0, \ldots, n-1\}\}. \tag{8}$$

The Variational Principle yields

$$\lim_{n \to +\infty} \frac{1}{n} \log \sum_{x_0, \ldots, x_{n-1} \in \{0,1\}} \sup_{\underline{x} \in [x_0 \ldots x_{n-1}]} \exp(S_n(\phi)(\underline{x})) = P_\sigma(\phi), \tag{9}$$

see, for example, [Bow08, Theorem 2.17].

Let ε in $(0,+\infty)$ be given. By uniform continuity, there is n_0 in $\mathbb{N}$ such that, for all x_0, $\ldots$, x_{n_0-1} in $\{0,1\}$ and $\underline{x}$ and $\underline{x}'$ in $[x_0\ldots x_{n_0-1}]$, the inequality $|\phi(\underline{x})-\phi(\underline{x}')|\leqslant\varepsilon$ holds. It follows that, for all n in $\mathbb{N}$ satisfying $n\geqslant n_0$, x_0, $\ldots$, x_{n-1} in $\{0,1\}$, and $\underline{x}$ and $\underline{x}'$ in $[x_0\ldots x_n]$,

$$|S_n(\phi)(\underline{x})-S_n(\phi)(\underline{x}')|\leqslant(n-n_0)\varepsilon+2n_0\|\varphi\|_\Sigma. \tag{10}$$

Thus,

$$\begin{aligned}\mathcal{L}^n_{\sigma,\phi}\mathbb{1}_\Sigma(\underline{x})&\leqslant\sum_{x_0,\ldots,x_{n-1}\in\{0,1\}}\sup_{\underline{x}'\in[x_0\ldots x_{n-1}]}\exp(S_n(\phi)(\underline{x}'))\\&\leqslant\exp((n-n_0)\varepsilon+2n_0\|\varphi\|_\Sigma)\mathcal{L}^n_{\sigma,\phi}\mathbb{1}_\Sigma(\underline{x})\end{aligned}\tag{11}$$

and

$$\left|\frac{1}{n}\log\mathcal{L}^n_{\sigma,\phi}\mathbb{1}_\Sigma(\underline{x})-\frac{1}{n}\log\sum_{x_0,\ldots,x_{n-1}\in\{0,1\}}\sup_{\underline{x}'\in[x_0\ldots x_{n-1}]}\exp(S_n(\phi)(\underline{x}'))\right|\leqslant\frac{n-n_0}{n}\varepsilon+\frac{2n_0}{n}\|\varphi\|_\Sigma. \tag{12}$$

Together with (9), this yields the lemma. $\square$

Proof of Lemma 3. Put $I:=[0,1]$ and let $\widetilde{f}\colon\widetilde{I}\to\widetilde{I}$ be the continuous extension of f obtained by "doubling" the points of $\bigcup_{n=0}f^{-n}(x_1)$, as in [CIR25, Proof of Lemma 2.6]. Put $Y^+:=\bigcup_{n=0}^{+\infty}\widetilde{f}^{-n}(x_1^+)$, and denote by $\pi\colon\widetilde{I}\to I$ the corresponding projection and $\widetilde{\varphi}\colon\widetilde{I}\to\mathbb{R}$ the continuous extension of $\widehat{\varphi}|_{\widetilde{I}\setminus Y^+}$. Then, $\pi\circ\widetilde{f}=f\circ\pi$, $\widetilde{\varphi}=\widehat{\varphi}\circ\pi$, and $\pi\colon\widetilde{I}\setminus Y^+\to I$ is a Borel measurable bijection conjugating $\widetilde{f}|_{\widetilde{I}\setminus Y^+}$ to f.

Since $\widetilde{f}$ is topologically conjugate to the full shift in two symbols [CIR25, Lemma 2.6], the dual $\mathcal{L}^*_{\widetilde{f},\widetilde{\varphi}}$ of the transfer operator $\mathcal{L}_{\widetilde{f},\widetilde{\varphi}}$ of $\widetilde{f}$ for the potential $\widetilde{\varphi}$ is continuous and has an eigenprobability $\widetilde{m}$ by the Schauder–Tychonoff Fixed Point Theorem, see, for example, [Bow08, p.9]. Denoting by λ the egeinvalue of $\widetilde{m}$, it follows that, for every n in $\mathbb{N}$,

$$\lambda^n\widetilde{m}(\widetilde{I})=(\mathcal{L}^*_{\widetilde{f},\widetilde{\varphi}})^n\widetilde{m}(\widetilde{I})=\int\mathbb{1}_{\widetilde{I}}\,\mathrm{d}(\mathcal{L}^*_{\widetilde{f},\widetilde{\varphi}})^n\widetilde{m}=\int\mathcal{L}^n_{\widetilde{f},\widetilde{\varphi}}\mathbb{1}_{\widetilde{I}}\,\mathrm{d}\widetilde{m}. \tag{13}$$

Combined with Lemma 4, this yields

$$\log\lambda=\lim_{n\to+\infty}\frac{1}{n}\log\int\mathcal{L}^n_{\widetilde{f},\widetilde{\varphi}}\mathbb{1}_{\widetilde{I}}\,\mathrm{d}\widetilde{m}=P_{\widetilde{f}}(\widetilde{\varphi}) \tag{14}$$

and thus $\mathcal{L}^*_{\widetilde{f},\widetilde{\varphi}}\widetilde{m}=\exp(P_{\widetilde{f}}(\widetilde{\varphi}))\widetilde{m}$. It follows that $\widetilde{m}$ is $\exp(P(\beta\widetilde{\varphi})-\beta\widetilde{\varphi})$-conformal for $\widetilde{f}$. Since $\widetilde{f}$ is topologically exact on $\widetilde{I}$, the equality $\operatorname{supp}(\widetilde{m})=\widetilde{I}$ holds.

To prove that $\widetilde{m}$ is atom-free, let $\widetilde{x}$ in $\widetilde{I}$ be given. For every k in $\mathbb{N}$,

$$\begin{aligned}\widetilde{m}(\{x\})&=\exp(-(kP(\widehat{\varphi})-S_k(\widetilde{\varphi})(\widetilde{x})))\widetilde{m}(\{\widetilde{f}^k(x)\})\\&\leqslant\exp\left(-k\left(P(\widehat{\varphi})-\sup_{[0,1]}\widehat{\varphi}\right)\right)\widetilde{m}(\{\widetilde{f}^k(\widetilde{x})\}).\end{aligned}\tag{15}$$

Letting $k\to+\infty$ and using (4) in Lemma 2 yields $\widetilde{m}(\{\widetilde{x}\})=0$. This proves that $\widetilde{m}$ is atom-free.

Put $m:=\pi_*\widetilde{m}$. Since $\widetilde{m}$ is atom-free and $\exp(P(\beta\widetilde{\varphi})-\beta\widetilde{\varphi})$-conformal for $\widetilde{f}$, the measure m is atom-free and $(1/g)$-conformal for f. Moreover, $\widetilde{m}(\widetilde{I})=1$ and $\operatorname{supp}(\widetilde{m})=\widetilde{I}$ imply $m(I)=1$ and $\operatorname{supp}(m)=I$. $\square$

Proof of Proposition 1. Let β be in $(0, +\infty)$, put $g := \exp(\beta\widehat{\varphi} - P(\beta\widehat{\varphi}))$ as in Lemma 3, and put $\gamma_* := \min\{1, \gamma\}$. The functions $\widehat{\varphi}$ and g belong to $C^{\gamma_*}([0,1], \mathbb{R})$ and g satisfies $\sup_{[0,1]} g < 1$ by Lemma 2. Let m be the atom-free $(1/g)$-conformal probability measure for f satisfying $\mathrm{supp}(m) = [0,1]$ given by Lemma 3.

The first step is to verify the hypotheses of [LRL14, Theorem 1 in §4.2], which is a reformulation of results of Keller [Kel85, Theorems 3.2 and 3.3]. Hypotheses H1 and H2 of [LRL14, Theorem 1 in §4.2] follow automatically from the fact that every function in $C^{\gamma_*}([0,1], \mathbb{R})$ is of bounded $(1/\gamma_*)$-variation and that m is $(1/g)$-conformal for f. Hypothesis H3 of [LRL14, Theorem 1 in §4.2] can be shown as in the proof of [LRL14, Lemma 5.1]. Thus, the hypotheses of [LRL14, Theorem 1 in §4.2] are satisfied. Combined with the fact that f is topologically exact on $(0,1]$ by [CIR25, Lemma 2.8] and well-known arguments, this result implies that $t \mapsto P(t\varphi)$ is real-analytic at $t = \beta$. □

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